

Changyu Dong

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EDUCATION

PhD, Imperial College London	10.2005 – 02.2009
MSc (Distinction), Imperial College London	10.2004 – 10.2005
BSc, Fudan University	09.1995 – 09.1999

ACADEMIC AND PROFESSIONAL APPOINTMENTS

MoE Yangtze River Chair Professor, Guangzhou University	07.2022 – Present
Senior Lecturer, Newcastle University	04.2016 – 06.2022
Lecturer, University of Strathclyde	04.2011 – 04.2016
Research Assistant/Associate, Imperial College London	10.2005 – 04.2011
IT Supervisor, Shanghai Pan Pacific Express Ltd.	11.2001 – 08.2004
Systems Engineer, Shanghai Infosource Communication Corp.	09.2000 – 10.2001

RESEARCH PROFILE

Research focuses on trustworthy data-driven systems, with emphasis on applied cryptography, differential privacy, secure computation, and AI security. The work addresses privacy, robustness, and accountability in computation and machine learning over sensitive data. Contributions include privacy-preserving technologies for large-scale data analytics, federated and distributed learning, searchable encryption, private set intersection, blockchain-based trust, and secure AI. The research has informed practical systems and methods for privacy-preserving data use in real-world settings where sensitive information must be analyzed, shared, or learned from securely. Publications appear in leading venues in security, privacy, artificial intelligence, and IEEE/ACM journals and conferences, including USENIX Security, ACM CCS, NDSS, ICML, CVPR, IEEE TIFS, IEEE TDSC, IEEE TPDS, IEEE TKDE, and ACM TOPS.

SELECTED RESEARCH GRANTS

2025 – Present, **Beijing Zitiao Network Technology Co., Ltd.**, “TEE-assisted Secure Computation”, ¥400,000, PI
2023 – Present, **NSFC (62261160651)**, “Privacy-Preserved Cross-border Data Collaborative Management for Credit Investigation”, ¥1,650,000, co-PI
2024 – 2025, **Guangzhou Basic and Applied Basic Research Project (2024A03J0324)**, “Research on the AI Full Life Cycle Safety Assessment Framework”, ¥1,000,000, PI
2022 – 2025, **EPSRC (EP/W032481)**, “AGENCY: Assuring Citizen Agency in a World with Complex Online Harms”, £2,793,095, co-I
2021 – 2024, **NSFC (62072132)**, “Research on Inference Attack and Defence in Multiparty Machine Learning”, ¥560,000, PI
2020 – 2021, **EPSRC IAA**, “Defence Against Adversarial Examples”, £13,454, PI
2017 – 2020, **EPSRC (EP/M013561/1, EP/M013561/2)**, “Practical Data-intensive Secure Computation: A Data Structural Approach”, £462,480, PI
2016 – 2017, **Crypta Labs Ltd.**, “Statistical Tests for Quantum Random Number Generator (QRNG)”, £5,600, PI
2014, **SICSA PECE grant**, “Privacy Preserving Vehicle Networks”, £2,900, PI

PROFESSIONAL SERVICE

Conference Organisation

AIS&P 2023, 2025 (program committee chair): International Conference on Artificial Intelligence Security and Privacy

SPNCE 2019 (program committee chair): International Conference on Security and Privacy in New Computing Environments.

IEEE ICPADS 2014 (vice program committee chair): International Conference on Parallel and Distributed Systems

Prism 2013 (publicity chair): International Conference on Privacy and Security in Mobile Systems.

Program Committee Member

IEEE Symposium on Security and Privacy (**Oakland**) 2026,2027, USENIX Security Symposium (**USENIX Security**) 2024,2025,2026, European Symposium on Research in Computer Security (**ESORICS**) 2020, 2021, 2022, IEEE International Conference on Trust, Security and Privacy in Computing and Communications (**IEEE Trustcom**) 2019, 2020, 2021, 2021,2022,2023, IEEE International Conference on Blockchain (**IEEE Blockchain**) 2020, 2022,2023,2024, International Symposium on Dependability in Sensor, Cloud, and Big Data Systems and Applications (**DependSys**) 2015, 2016, 2017, 2018, International Conference on Information Systems Security and Privacy (**ICISSP**) 2015, 2016, IEEE International Workshop on Privacy Aspects of Data Mining (**IEEE PADM**) 2014, IEEE International Workshop on Engineering Energy Efficient Wireless Sensor Networks (**IEEE EEEWSN**) 2014, 2015, 2018, International Conference on Autonomic and Autonomous Systems (**ICAS**) 2014, Scalable Cloud Data Management Workshop (**SCDM**) 2013, 2014, 2015, International Conference on Emerging Security Information, Systems and Technologies (**SecureWare**), 2013, 2014, 2015,2016,2017

SELECTED PUBLICATIONS

1. Shuyan Zou, Shaowei Wang, Zhanxing Zhu, Jin Li, **Changyu Dong**, Vladimiro Sassone, and Han Wu, SlaClip: Gradient Norm Slacks can be Indicator for Adaptive Clipping in DP-SGD, in Proceedings of 43rd International Conference on Machine Learning (ICML) (2026). **Spotlight Paper**
2. Shaoqiang Wu, Jingyu Jia, Yikuan Zhu, Xinhao Li, **Changyu Dong**, and Zheli Liu, High-Accuracy, Poisoning-Resilient Frequency Estimation in the Shuffle Model, in Proceedings of 35th USENIX Security Symposium (2026).
3. Teng Huang, Jiahui Huang, **Changyu Dong**, Sisi Duan, and Yan Pang, SAMamba: Structure-Aware Mamba for Ethereum Fraud Detection, IEEE Trans. Inf. Forensics Secur. (2025).
4. Xingyu He, Youwen Zhu, Rongke Liu, Gaoning Pan, and **Changyu Dong**, Addressing Sensitivity Distinction in Local Differential Privacy: A General Utility-Optimized Framework, in Proceedings of 34th USENIX Security Symposium (2025).
5. Jiahui Huang, Teng Huang, **Changyu Dong**, Sisi Duan, and Yan Pang, Hierarchical Network With Local-Global Awareness for Ethereum Account De-anonymization, IEEE Transactions on Systems, Man, and Cybernetics: Systems 1-13 (2025).
6. Songze Li, Jiaxiang Tang, Jinbao Zhu, Kai Zhang, Lichao Sun, and **Changyu Dong**, Secure Embedding Aggregation for Cross-Silo Federated Representation Learning, IEEE Trans. Inf. Forensics Secur. 1-13 (2025).
7. Shaowei Wang, Jin Li, **Changyu Dong**, Jin Li, Zhili Zhou, Di Wang, and Zikai Wen, Side-channel Attacks and New Principles in the Shuffle Model of Differential Privacy, IEEE Trans. Inf. Forensics Secur. 1-13 (2025).
8. Shaowei Wang, **Changyu Dong**, Xiangfu Song, Jin Li, Zhili Zhou, Di Wang, and Han Wu, Beyond Statistical Estimation: Differentially Private Individual Computation via Shuffling, in Proceedings of 34th USENIX Security Symposium (2025). **Honorable Mentions Award**
9. Zuopeng Yang, Jiluan Fan, Anli Yan, Erdun Gao, Xin Lin, Tao Li, Kanghua Mo, and **Changyu Dong**, Distraction is All You Need for Multimodal Large Language Model Jailbreaking, in Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2025). **Spotlight Paper**
10. Lei Zhang, Yazhou Ma, Mingzi Zuo, Zhen Ling, **Changyu Dong**, Guangquan Xu, Lin Shu, Xiaochen Fan, and Qian Zhang, Toward Cross-Environment Continuous Gesture User Authentication With Commercial Wi-Fi, IEEE Transactions on Networking 1-13 (2025).
11. Yaxi Yang, Xiaojian Liang, Xiangfu Song, Ye Dong, Linting Huang, Hongyu Ren, **Changyu Dong**, and Jianying Zhou, Maliciously Secure Circuit Private Set Intersection via SPDZ-Compatible Oblivious PRF,

- in Proceedings of 25th Privacy Enhancing Technologies Symposium (PETS) (2025).
12. Lei Zhang, Yunzhe Jiang, Yazhou Ma, Shiwen Mao, Wenyan Huang, Zhiyong Yu, Xiao Zheng, Lin Shu, Xiaochen Fan, Guangquan Xu, and **Changyu Dong**, Toward Robust and Effective Behavior Based User Authentication With Off-the-Shelf Wi-Fi, *IEEE Trans. Inf. Forensics Secur.* 19: 8731–8746 (2024).
 13. Jingyu Jia, Xinhao Li, Tong Li, Zhewei Liu, Chang Tan, Siyi Lv, Liang Guo, **Changyu Dong**, and Zheli Liu, ABSyn: An Accurate Differentially Private Data Synthesis Scheme With Adaptive Selection and Batch Processes, *IEEE Trans. Inf. Forensics Secur.* 19: 8338–8352 (2024).
 14. Xiangfu Song, Dong Yin, Jianli Bai, **Changyu Dong**, and Ee-Chien Chang, Secret-Shared Shuffle with Malicious Security, in Proceedings of 31st Annual Network and Distributed System Security Symposium (NDSS) (2024).
 15. Liqun Chen, **Changyu Dong**, Christopher J. P. Newton, and Yalan Wang, Sphinx-in-the-Head: Group Signatures from Symmetric Primitives, *ACM Trans. Priv. Secur.* 27(1): 11:1–11:35 (2024).
 16. Yu Wei, Jingyu Jia, Yuduo Wu, Changhui Hu, **Changyu Dong**, Zheli Liu, Xiaofeng Chen, Yun Peng, and Shaowei Wang, Distributed Differential Privacy via Shuffling Versus Aggregation: A Curious Study, *IEEE Trans. Inf. Forensics Secur.* 19: 2501–2516 (2024).
 17. Boyu Zhu, **Changyu Dong**, Yuan Zhang, Yunlong Mao, and Sheng Zhong, Toward Universal Detection of Adversarial Examples via Pseudorandom Classifiers, *IEEE Trans. Inf. Forensics Secur.* 19: 1810–1825 (2024).
 18. Yi Xie, Mengdie Huang, Xiaoyu Zhang, **Changyu Dong**, Willy Susilo, and Xiaofeng Chen, GAME: Generative-Based Adaptive Model Extraction Attack, in Proceedings of 27th European Symposium on Research in Computer Security Computer Security (ESORICS) (2022).
 19. Li Hu, Jin Li, Guanbiao Lin, Shiyu Peng, Zhenxin Zhang, Yingying Zhang, and **Changyu Dong**, Defending against Membership Inference Attacks with High Utility by GAN, *IEEE Trans. Dependable Secur. Comput.* (2022).
 20. Xu Ma, Xiaoqian Sun, Yuduo Wu, Zheli Liu, Xiaofeng Chen, and **Changyu Dong**, Differentially Private Byzantine-robust Federated Learning, *IEEE Trans. Parallel Distributed Syst.* (2022).
 21. Huiping Chen, **Changyu Dong**, Liyue Fan, Grigorios Loukides, Solon P. Pissis, and Leen Stougie, Differentially Private String Sanitization for Frequency-Based Mining Tasks, in Proceedings of IEEE International Conference on Data Mining (ICDM) (2021).
 22. Changhui Hu, Jin Li, Zheli Liu, Xiaojie Guo, Yu Wei, Xuan Guang, Grigorios Loukides, and **Changyu Dong**, How to Make Private Distributed Cardinality Estimation Practical, and Get Differential Privacy for Free, in Proceedings of 30th USENIX Security Symposium (2021).
 23. Xiaojie Guo, Zheli Liu, Jin Li, Jiqiang Gao, Boyu Hou, **Changyu Dong**, and Thar Baker, VeriFL: Communication-Efficient and Fast Verifiable Aggregation for Federated Learning, *IEEE Trans. Inf. Forensics Secur.* 16: 1736–1751 (2021).
 24. Jin Li, Yanyu Huang, Yu Wei, Siyi Lv, Zheli Liu, **Changyu Dong**, and Wenjing Lou, Searchable Symmetric Encryption with Forward Search Privacy, *IEEE Trans. Dependable Secur. Comput.* 18(1): 460–474 (2021).
 25. Zheli Liu, Yanyu Huang, Xiangfu Song, Bo Li, Jin Li, Yali Yuan, and **Changyu Dong**, Euris: Towards an Efficient Searchable Symmetric Encryption with Size Pattern Protection, *IEEE Trans. Dependable Secur. Comput.* (2020).
 26. Zheli Liu, Jin Li, Siyi Lv, Yanyu Huang, Liang Guo, Yali Yuan, and **Changyu Dong**, EncodeORE: Reducing Leakage and Preserving Practicality in Order-Revealing Encryption, *IEEE Trans. Dependable Secur. Comput.* (2020).
 27. Chongzhi Gao, Jin Li, Shibing Xia, Kim-Kwang Raymond Choo, Wenjing Lou, and **Changyu Dong**, MAS-Encryption and Its Applications in Privacy-Preserving Classifiers, *IEEE Trans. Knowl. Data Eng.* (2020).
 28. Xiangfu Song, **Changyu Dong**, Dandan Yuan, Qiuliang Xu, and Minghao Zhao, Forward Private Searchable Symmetric Encryption with Optimized I/O Efficiency, *IEEE Trans. Dependable Secur. Comput.* 17(5): 912–927 (2020).
 29. Aydin Abadi, Sotirios Terzis, Roberto Metere, and **Changyu Dong**, Efficient Delegated Private Set In-

- tersection on Outsourced Private Datasets, *IEEE Trans. Dependable Secur. Comput.* 16(4): 608–624 (2019).
30. Feng Hao, Roberto Metere, Siamak F. Shahandashti, and **Changyu Dong**, Analyzing and Patching SPEKE in ISO/IEC, *IEEE Trans. Inf. Forensics Secur.* 13(11): 2844–2855 (2018).
 31. **Changyu Dong** and Grigorios Loukides, Approximating Private Set Union/Intersection Cardinality With Logarithmic Complexity, *IEEE Trans. Inf. Forensics Secur.* 12(11): 2792–2806 (2017).
 32. **Changyu Dong**, Yilei Wang, Amjad Aldweesh, Patrick McCorry, and Aad van Moorsel, Betrayal, Distrust, and Rationality: Smart Counter-Collusion Contracts for Verifiable Cloud Computing, in *Proceedings of ACM SIGSAC Conference on Computer and Communications Security (CCS)* (2017).
 33. **Changyu Dong** and Liqun Chen, A Fast Single Server Private Information Retrieval Protocol with Low Communication Cost, in *Proceedings of 19th European Symposium on Research in Computer Security* (2014).
 34. **Changyu Dong**, Liqun Chen, and Zikai Wen, When private set intersection meets big data: an efficient and scalable protocol, in *Proceedings of ACM SIGSAC Conference on Computer and Communications Security (CCS)* (2013).